

Homework # 2: Bayes, admissible, and minimax decision rules. Prior distribution.*Due February 7*

1. (Berger, 1.1) Let X have Poisson(θ) distribution, $\Theta = (0, \infty)$, and $\mathcal{A} = [0, \infty)$. The loss function is $L(\theta, \delta) = (\theta - \delta)^2$. Assume $\pi(\theta) = e^{-\theta}$ is the prior density.
 - (a) First, assume X is not observed. Calculate the *prior risk* $\rho(\pi, \delta) = \mathbf{E}^\pi L(\theta, \delta)$ of a decision δ and find the Bayes decision.
 - (b) Now, suppose X is observed, and from now on, we use decision rules of the form $\delta_c(x) = cx$. Find the frequentist risk $R(\theta, \delta_c)$.
 - (c) Show that δ_c is inadmissible if $c > 1$.
 - (d) Find the Bayes risk of δ_c .
 - (e) Find the value of c which minimizes the Bayes risk.
 - (f) Is there a best rule of the form δ_c in terms of the minimax principle?
2. (Berger 1.2) An insurance company is faced with taking one of the following 3 decisions:

- $$\begin{aligned}
 a_1 &= \text{increase sales force by 10\%} \\
 a_2 &= \text{maintain present sales force} \\
 a_3 &= \text{decrease sales force by 10\%}
 \end{aligned}$$

Depending upon whether or not the economy is good (θ_1), mediocre (θ_2), or bad (θ_3), the company would expect to lose the following amounts of money in each case:

		Decision taken		
		a_1	a_2	a_3
State of Economy	θ_1	-10	-5	-3
	θ_2	-5	-5	-2
	θ_3	1	0	-1

- (a) Determine if each decision is admissible or inadmissible.
 - (b) The company believes that θ has the probability distribution $\pi(\theta_1) = 0.2$, $\pi(\theta_2) = 0.3$, $\pi(\theta_3) = 0.5$. Order the decisions according to their Bayes risk and state the Bayes action.
 - (c) Order the decisions according to the minimax principle and find the minimax decision.
3. (Berger 1.5) A farmer has to decide whether or not to plant his crop early. If he plants early and no late frost occurs, he will gain \$5000 in extra harvest. If he plants early and a late frost does occur, he will lose \$2000 as the cost of reseeding. If he doesn't plant early, his gain will be \$0. Consulting the weather service, he finds that the chance of late frost is 0.6.
 - (a) Describe $\mathcal{A}$, Θ , the loss function, and the prior distribution.
 - (b) What is the Bayes decision?
 - (c) What is the minimax (nonrandomized) decision?

4. (Berger, 3.2) Let θ denote the highest temperature that will occur outdoors tomorrow, near your place of residence. (If you'd like, take your place of birth instead.) Using the histogram approach, find your subjective prior density for θ . Briefly explain the steps.
5. (Berger, 3.4) For the parameter θ in the previous exercise,
 - (a) Determine the 1st and 2nd quartiles of your prior density for θ .
 - (b) Find the normal density matching these quartiles (find the corresponding parameters μ and σ).
 - (c) Find, *subjectively*, the 10th and the 33rd percentiles of your prior distribution for θ . (Do not use the Normal distribution from [b] to obtain these.) Are these percentiles consistent with the Normal distribution in [b]? Is the Normal density a good fit for your prior density of θ ? (After all, $\pi(\theta)$ does not have to be Normal.)

For definitions of quartiles and percentiles, see Sec. 3.1.3 of our lecture notes.

6. A random sample of 10 students yields the average weight of 150 pounds. Assume the weights in the population are normally distributed with unknown mean θ and known standard deviation 20 pounds. Suppose the prior distribution of θ is Normal with mean 180 and standard deviation 40. Determine the posterior distribution of θ , the population average weight; specify the posterior mean and posterior standard deviation.
7. For a sample from Uniform(0, θ) distribution, propose a conjugate family of prior distributions and prove that it is conjugate to the Uniform model.
8. (Berger, 3.8) For each of the following densities, state whether it represents a location family or a scale family and give the natural non-informative prior for the unknown parameter.
 - (a) Uniform($\theta - 1, \theta + 1$),
 - (b) Cauchy(0, β) with a density $f(x | \beta) = \frac{\beta}{\pi(\beta^2 + x^2)}$, $x \in (-\infty, \infty)$.
 - (c) Pareto(2, σ) with a density $f(x | 2, \sigma) = 2\sigma^2 x^{-3}$, $x > \sigma$
(it is in our inventory of distributions).